

04.04 Operations with Complex Numbers

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
Adding or subtracting complex numbers.	Combine real terms and imaginary terms separately.
Multiplying complex numbers.	Distribute and replace every i^2 with -1 .
Dividing by a complex number.	Multiply numerator and denominator by the denominator's conjugate.

A. Add or subtract

1.

$$(3 + 2i) + (5 - 7i)$$

2.

$$(6 - 4i) + (2 + 9i)$$

3.

$$(8 + i) - (3 - 5i)$$

4.

$$(1 - 6i) + (7 + 2i)$$

5.

$$(9 + 3i) - (4 + i)$$

6.

$$(2 + 8i) + (5 - 3i)$$

7.

$$(7 - 2i) - (1 + 4i)$$

8.

$$(4 + 5i) + (6 - 9i)$$

9.

$$(10 - i) - (3 + 2i)$$

10.

$$(5 + 7i) + (2 - 8i)$$

11.

$$(11 - 3i) - (6 - 5i)$$

12.

$$(2 - i) + (9 + 6i)$$

13.

$$(8 + 4i) - (2 - 7i)$$

14.

$$(1 + 9i) + (3 - 2i)$$

15.

$$(12 - 5i) - (4 + 3i)$$

16.

$$(7 + 2i) + (1 + 8i)$$

17.

$$(6 - 7i) - (2 - i)$$

18.

$$(3 + 4i) + (8 - 6i)$$

19.

$$(9 - 2i) - (5 + 5i)$$

20.

$$(4 + 6i) + (7 - 3i)$$

B. Multiply

1.

$$(3 + 2i)(1 - 4i)$$

2.

$$(2 - i)(5 + 3i)$$

3.

$$(4 + 3i)(4 - 3i)$$

4.

$$(1 + 5i)(2 - i)$$

5.

$$(6 - 2i)(1 + 4i)$$

6.

$$(3 - i)^2$$

7.

$$(2 + 5i)^2$$

8.

$$(7 + 2i)(7 - 2i)$$

9.

$$(1 - 3i)(4 + 2i)$$

10.

$$(5 + i)(2 - 6i)$$

11.

$$(2 - 4i)(3 - 5i)$$

12.

$$(8 + i)(1 - 2i)$$

13.

$$(4 - 7i)(2 + i)$$

14.

$$(3 + 6i)(3 - 6i)$$

15.

$$(1 + 2i)^2$$

16.

$$(5 - 2i)(-1 + 3i)$$

17.

$$(2 + 3i)(-4 + i)$$

18.

$$(6 + i)(2 + i)$$

19.

$$(1 - 4i)(1 + 4i)$$

20.

$$(3 - 2i)^2$$

C. Divide using conjugates

1.

$$\frac{4 + 3i}{2 - i}$$

2.

$$\frac{5 - i}{1 + 2i}$$

3.

$$\frac{3 + 4i}{2 + i}$$

4.

$$\frac{7 - 2i}{3 - i}$$

5.

$$\frac{1 + 5i}{4 + i}$$

6.

$$\frac{6 - i}{2 - 3i}$$

7.

$$\frac{8 + 2i}{1 - i}$$

8.

$$\frac{9 - 3i}{2 + i}$$

9.

$$\frac{2 + 7i}{5 - i}$$

10.

$$\frac{4 - 5i}{3 + 2i}$$

11.

$$\frac{1 - 2i}{2 - i}$$

12.

$$\frac{6 + 5i}{4 - i}$$

13.

$$\frac{7 + i}{1 + 3i}$$

14.

$$\frac{3 - 6i}{2 + 2i}$$

15.

$$\frac{5 + 4i}{3 - i}$$

16.

$$\frac{2 - i}{1 - 4i}$$

17.

$$\frac{8+i}{5+2i}$$

18.

$$\frac{9-i}{2-5i}$$

19.

$$\frac{4+6i}{3+i}$$

20.

$$\frac{7-4i}{1-2i}$$